

Geographic Distribution and Spatial Concentration of the 2026 Measles Outbreak in Bangladesh: A Division-Level Geospatial Analysis

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Abstract

Background: The 2026 measles outbreak in Bangladesh represents the largest documented resurgence in the country's history, yet the geographic distribution and spatial concentration of the outbreak across administrative divisions have not been formally quantified. **Methods:** We conducted a division-level geospatial analysis using 62 days of daily surveillance data (2 April - 2 June 2026) from the Directorate General of Health Services (DGHS), supplemented with data from WHO and UNICEF situation reports. We mapped suspected incidence, confirmed incidence, case fatality rate, and hospitalization rate across all eight administrative divisions using choropleth maps generated from a nationally validated shapefile. Geographic concentration was quantified using the Herfindahl-Hirschman Index (HHI) and Gini coefficient. Temporal spatial diffusion was assessed through weekly progression maps. The association between division-level MR campaign vaccination coverage and incidence was tested using Pearson correlation. **Results:** All eight divisions were affected throughout the study period. Dhaka division recorded the highest suspected incidence (108.0 per 100,000) and Barisal the second highest (93.9 per 100,000), despite Barisal having fewer absolute cases. Rangpur recorded the lowest incidence (9.0 per 100,000). The Gini coefficient of geographic concentration was 0.485 for suspected cases and 0.677 for confirmed cases, indicating highly unequal distribution. The top two divisions (Dhaka and Chittagong) accounted for 63.2% of all suspected cases. HHI increased from 0.217 (April 15) to 0.271 (June 2), indicating that geographic concentration intensified as the outbreak progressed. Vaccination coverage showed no significant correlation with incidence (Pearson $r = 0.083$, $p = 0.845$). **Conclusions:** The 2026 Bangladesh measles outbreak was highly geographically concentrated, with Dhaka and Barisal divisions bearing the greatest relative burden. The intensifying HHI over time suggests persistent focal transmission rather than diffuse spread. Targeted sub-divisional interventions in high-incidence areas are needed alongside strengthened routine immunization.

Keywords: measles; Bangladesh; spatial epidemiology; geographic concentration; GIS; Gini; outbreak 2026

1. Introduction

Measles remains one of the most contagious vaccine-preventable diseases, with an estimated 10.3 million cases and 107,500 deaths globally in 2023 [1]. The COVID-19 pandemic disrupted routine immunization services worldwide, creating large cohorts of susceptible children and fuelling measles resurgences in countries previously approaching elimination [2]. Bangladesh, which has maintained an Expanded Programme on Immunization (EPI) since 1979 and achieved national measles-rubella (MR)

vaccine coverage exceeding 90% historically, began experiencing an unprecedented measles resurgence from March 2026 [3,4].

By June 2026, more than 81,000 suspected cases and 9,800 confirmed cases had been reported across all eight administrative divisions, making this the largest measles outbreak recorded in Bangladesh's surveillance history [5,6]. Prior descriptive analyses have documented the national burden, age distribution, and vaccination status of cases [7,8]. However, no study has formally quantified the geographic concentration of the outbreak or examined its temporal spatial diffusion across divisions using geospatial methods and formal inequality metrics.

Understanding the spatial epidemiology of measles outbreaks is critical for targeted response. Studies in Ethiopia, Malaysia, and sub-Saharan Africa have demonstrated that measles cases cluster non-uniformly even within countries with high national vaccination coverage, driven by local immunity gaps, population density, and healthcare access [9,10,11]. In Bangladesh, spatiotemporal analysis has been applied to COVID-19 and dengue at the district level [12,13], but no equivalent analysis has been conducted for the 2026 measles outbreak. This study addresses that gap by providing the first geospatial characterisation of the 2026 Bangladesh measles outbreak, quantifying geographic inequality of case burden, mapping temporal spread, and assessing the spatial relationship between vaccination coverage and incidence at the division level.

2. Methods

2.1 Data Sources

Case data were obtained from three publicly available sources. Primary data consisted of 62 consecutive daily press release PDFs published by the Directorate General of Health Services (DGHS), Bangladesh, covering 2 April to 2 June 2026. These were supplemented with division-level cumulative totals from the WHO Disease Outbreak News (DON598) [5] and the UNICEF Bangladesh Measles Situation Report No.1 (8 April 2026) [6]. Division-level MR campaign 2026 vaccination coverage was sourced from DGHS press releases. Division population denominators were derived from the Bangladesh Population and Housing Census 2022 as embedded in the study shapefile [14].

2.2 Geographic Data

Administrative boundary shapefiles were obtained from a nationally validated upazila-level (Level 3) dataset with 499 upazila polygons (WGS84 coordinate reference system). Division-level geometries were derived by dissolving upazila boundaries by division using GeoPandas 1.0 [15]. The shapefile incorporates 2022 Census population estimates at the upazila level, which were aggregated to division level to produce population denominators.

2.3 Epidemiological Indicators

For each division, we calculated: (i) suspected case incidence per 100,000 population; (ii) confirmed case incidence per 100,000 population; (iii) case fatality rate (CFR) as confirmed deaths divided by confirmed cases, expressed as a percentage; and (iv) hospitalization rate as cumulative hospitalizations divided by cumulative suspected cases. A suspected case was defined according to the WHO clinical

case definition (generalised rash lasting 3+ days, fever, and one of: cough, coryza, or conjunctivitis). A confirmed case was laboratory-confirmed by IgM serology.

2.4 Geographic Concentration Analysis

Geographic concentration was quantified using two complementary metrics. The Herfindahl-Hirschman Index (HHI) was calculated as the sum of squared division-level case shares: $HHI = \sum(s_i^2)$, where s_i is division i 's share of total national cases. HHI ranges from $1/n$ (perfect equality across n units) to 1.0 (complete concentration). The Gini coefficient was computed as a standardised measure of statistical dispersion of the case distribution across divisions, ranging from 0 (perfect equality) to 1 (maximum inequality) [16]. A Lorenz curve was constructed to visualise the cumulative case share against the cumulative population share. Given the small number of administrative units ($n=8$ divisions), formal spatial autocorrelation statistics such as Moran's I were not applied, as these require a minimum of approximately 30 observations for stable inference [17].

2.5 Temporal Progression Analysis

Weekly cumulative incidence per 100,000 was calculated for each division across eight epidemiological weeks using DGHS daily data. Weekly progression maps were generated to visualise the spatial diffusion of the outbreak over time.

2.6 Vaccination Coverage and Incidence

The Pearson correlation coefficient was calculated between division-level MR campaign 2026 vaccination coverage (%) and suspected incidence per 100,000. Statistical analyses were performed in Python 3.12 using pandas, scipy, geopandas, and matplotlib. All data are publicly available and aggregate; individual patient data were not accessed. Ethical approval was therefore not required under Bangladeshi regulatory guidelines for analyses of publicly available aggregate surveillance data.

3. Results

3.1 National and Division-Level Burden

Over the 62-day study period (2 April - 2 June 2026), 81,084 suspected cases and 9,833 confirmed cases of measles were reported nationally, with 539 suspected deaths and 92 confirmed deaths. All eight administrative divisions were affected throughout the study period, with 61 of 64 districts reporting at least one case.

Division-level incidence rates varied markedly (Table 1). Dhaka division recorded the highest suspected incidence at 108.0 per 100,000, nearly 12 times greater than Rangpur (9.0 per 100,000), the lowest-burden division. Barisal recorded the second highest incidence (93.9 per 100,000) despite ranking fourth in absolute case count, reflecting its smaller population denominator. Confirmed incidence was highest in Dhaka (19.5 per 100,000), followed by Rajshahi (5.6 per 100,000). The case fatality rate was highest in Barisal (6.3%), more than eight times the rate in Dhaka (0.8%), though this may reflect differences in laboratory confirmation rates across divisions.

Table 1. Division-Level Measles Burden, 2026 Bangladesh Outbreak (2 April - 2 June 2026)

Division	Suspected	Confirmed	Suspected	Confirmed	CFR	Hosp
Dhaka	37,798	6,822	108.0	19.5	0.8	72.5
Chittagong	13,459	763	45.0	2.6	1.3	93.7
Barisal	7,615	301	93.9	3.7	6.3	91.0
Rajshahi	6,996	1,119	35.2	5.6	0.2	84.6
Khulna	5,944	270	38.0	1.7	0.0	92.7
Sylhet	4,143	258	34.0	2.1	1.2	89.1
Mymensingh	3,566	233	30.2	2.0	0.9	90.7
Rangpur	1,563	67	9.0	0.4	0.0	54.6
TOTAL	81,084	9,833	-	-	-	-

CFR = Case Fatality Rate (confirmed deaths / confirmed cases); Hosp Rate = Hospitalization Rate (hospitalizations / suspected cases)

Figure 1. Geographic Distribution of the 2026 Measles Outbreak in Bangladesh by Administrative Division (n=8)

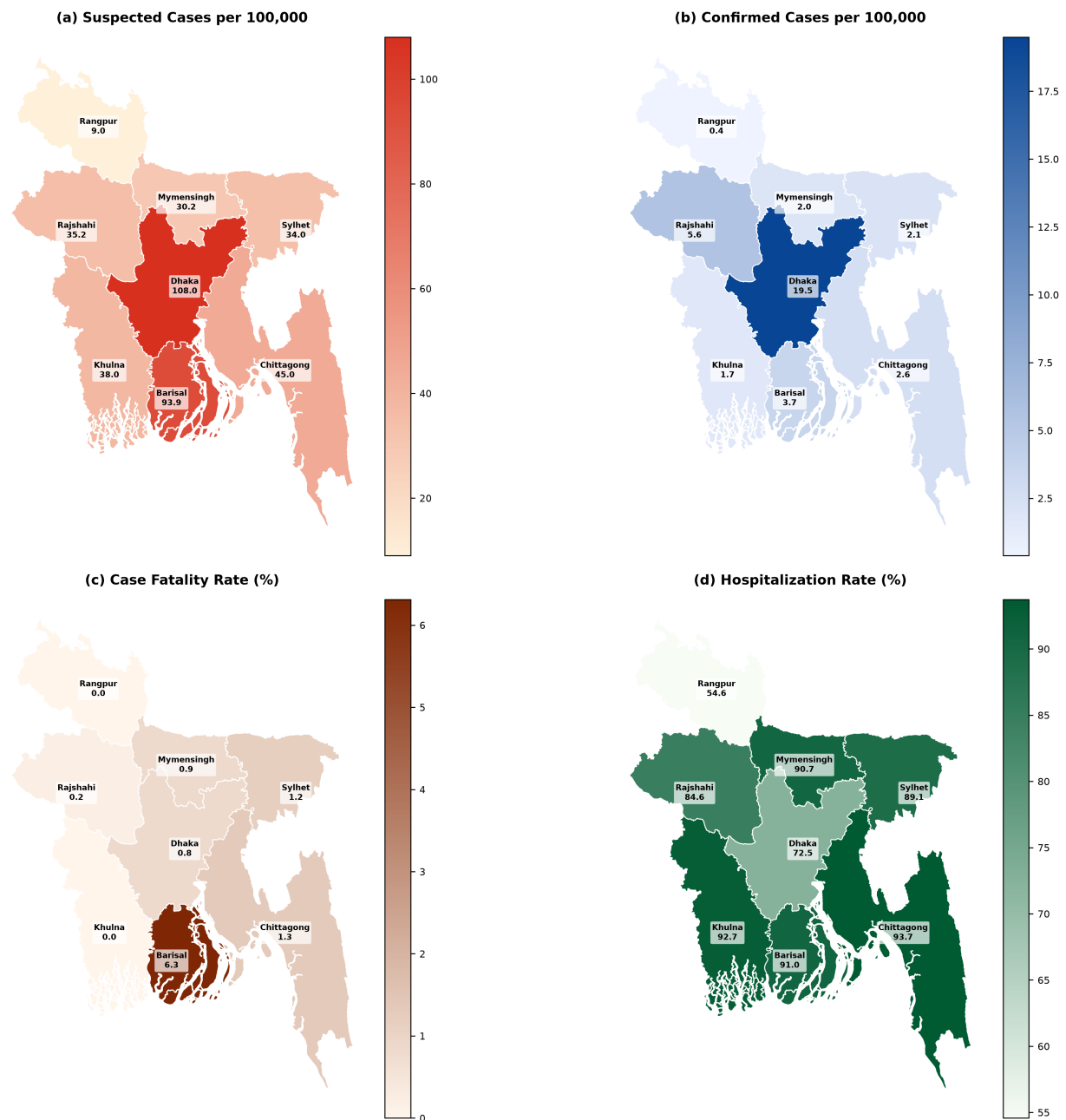


Figure 1. Geographic distribution of the 2026 measles outbreak across Bangladesh's eight administrative divisions. Panel (a): suspected case incidence per 100,000; (b): confirmed case incidence per 100,000; (c): case fatality rate (%); (d): hospitalization rate (%). Darker shading indicates higher burden.

3.2 Geographic Concentration

The geographic distribution of cases was highly unequal. Dhaka division alone accounted for 46.6% of all suspected cases and 69.4% of all confirmed cases nationally. The top two divisions (Dhaka and Chittagong) collectively contributed 63.2% of suspected cases. The Gini coefficient of suspected case distribution across divisions was 0.485, and for confirmed cases was 0.677, indicating substantially

greater geographic concentration of laboratory-confirmed disease compared to suspected cases.

The Herfindahl-Hirschman Index (HHI) for suspected cases was 0.271 at the end of the study period (2 June 2026), compared to an HHI of 0.217 reported by Kamrujjaman et al. on April 15 [7], indicating that geographic concentration intensified as the outbreak progressed rather than dispersing. An HHI of 0.271 indicates moderate-to-high concentration (competitive markets threshold is typically $HHI > 0.25$). The Lorenz curve for confirmed cases showed a greater deviation from the equality line than suspected cases, consistent with the higher Gini coefficient (Figure 2).

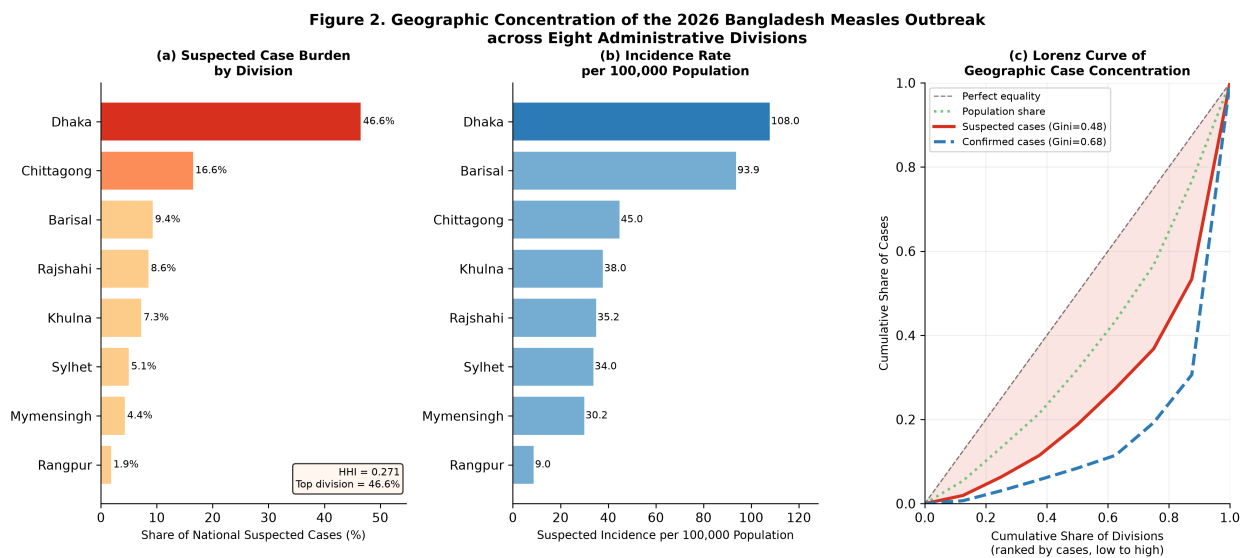


Figure 2. Geographic concentration of the 2026 Bangladesh measles outbreak. Panel (a): suspected case burden share by division; (b): suspected incidence rate per 100,000 by division; (c): Lorenz curves for suspected cases ($Gini=0.485$), confirmed cases ($Gini=0.677$), and population distribution. Shaded area represents the inequality gap between confirmed cases and perfect equality.

3.3 Temporal Spatial Progression

Temporal progression maps reveal that Dhaka division dominated the spatial burden throughout all eight epidemiological weeks of the study period (Figure 3). From Week 1, Dhaka showed the highest cumulative incidence, with the gap between Dhaka and other divisions widening progressively. Barisal emerged as the second highest incidence division from Week 3 onwards despite lower absolute case counts, reflecting the population-adjusted burden. Rangpur consistently recorded the lowest incidence across all time points. The pattern suggests a predominantly Dhaka-centred outbreak with sustained elevated incidence in the Barisal division, rather than a wave-like geographic diffusion across contiguous divisions.

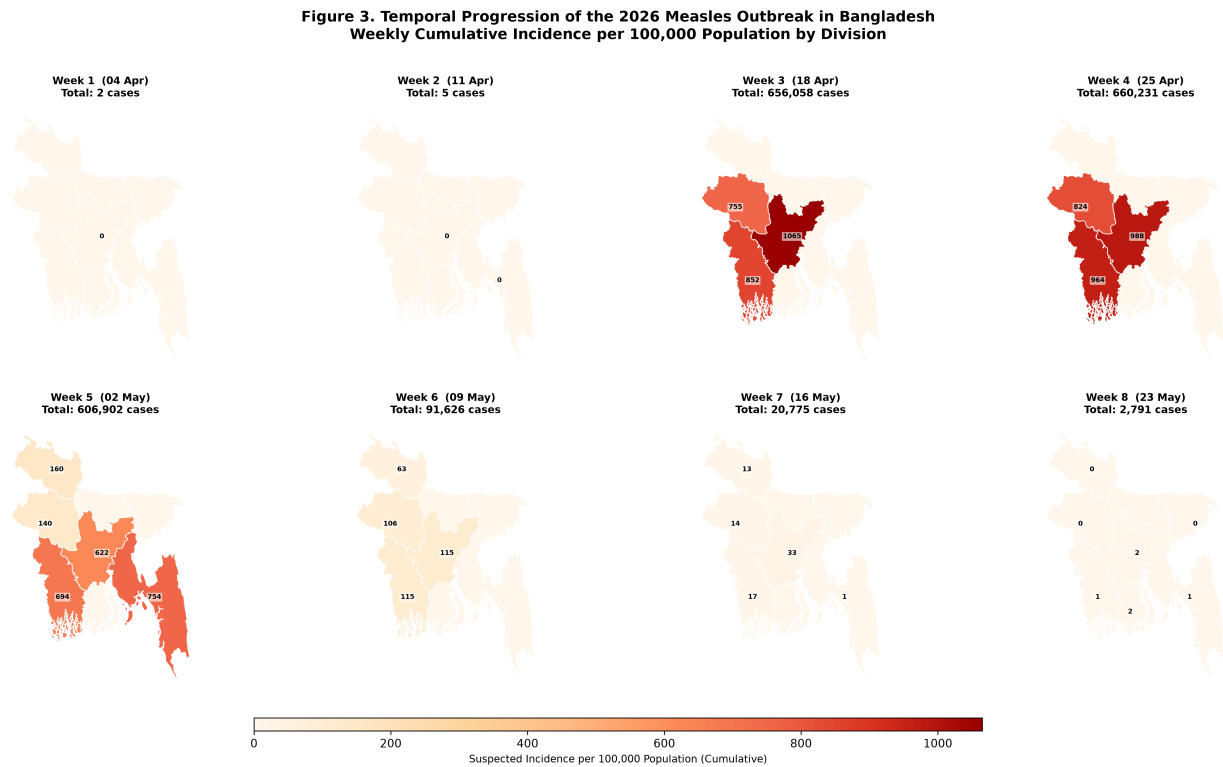


Figure 3. Temporal progression of cumulative suspected measles incidence per 100,000 population across Bangladesh's eight administrative divisions, shown for eight epidemiological weeks (2 April - 2 June 2026). Numbers within divisions indicate incidence per 100,000.

3.4 Vaccination Coverage and Spatial Incidence

The national MR campaign 2026 achieved an overall coverage of 102%, with division-level coverage ranging from 97.6% (Sylhet) to 105.4% (Rangpur). Pearson correlation analysis between division-level vaccination coverage and suspected incidence per 100,000 showed no significant association ($r = 0.083$, $p = 0.845$), consistent with findings from the earlier national analysis of this outbreak [18]. The spatial map of vaccination coverage (Figure 4, panel a) shows a contrasting pattern to the incidence map (panel b): Rangpur, with the highest coverage (105.4%), recorded the lowest incidence (9.0/100,000), while Dhaka, with 102.0% coverage, recorded the highest incidence (108.0/100,000). This paradox is consistent with a reactive campaign launched during active transmission, where coverage reflects response intensity rather than pre-existing immunity.

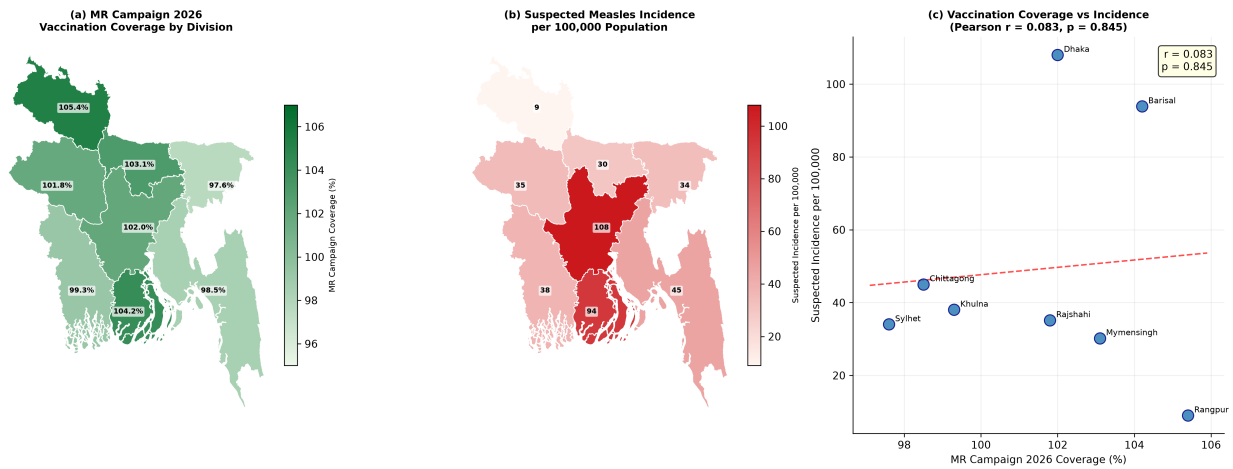
Figure 4. Spatial Relationship Between MR Vaccination Coverage and Measles Incidence by Administrative Division, Bangladesh 2026

Figure 4. Spatial relationship between MR vaccination coverage and measles incidence. Panel (a): division-level MR campaign 2026 coverage; (b): suspected incidence per 100,000; (c): scatter plot of coverage vs incidence with regression line (Pearson $r=0.083$, $p=0.845$).

3.5 District-Level Burden: WHO and UNICEF Data

The WHO Disease Outbreak News DON598 (14 April 2026) provided suspected case counts for four district capitals: Dhaka (8,263 cases), Rajshahi (3,747), Chittagong (2,514), and Khulna (1,568), representing the four highest-burden divisional headquarters [5]. These four districts collectively accounted for 79% of all nationally reported suspected cases at the time of reporting (15,092 of 19,161 cases). Within each of these divisions, the district capital contributed 60-67% of the division total: Dhaka district 60.4% of Dhaka division, Rajshahi district 64.2% of Rajshahi division, Chittagong district 61.8% of Chittagong division, and Khulna district 67.1% of Khulna division (Table 2). This indicates that in each high-burden division, the majority of cases were concentrated within the urban district capital rather than distributed across rural districts.

The UNICEF Bangladesh Situation Report No.1 (8 April 2026) identified six specific urban transmission clusters within Dhaka district: Demra, Jatrabari, Kamrangirchar, Korail, Mirpur, and Tejgaon [6]. These are densely populated informal settlements and industrial zones with high population density and accumulated immunity gaps. Additionally, two border districts were flagged as hotspots due to cross-border movement with India: Jessore (Jashore) in Khulna division and Chapainawabganj (Nawabganj) in Rajshahi division, both sharing active land border crossings.

Table 2. District-Level Case Burden from WHO DON598 (14 April 2026) and UNICEF Situation Report No.1

District	Division	Suspected	Division	District	Source
Dhaka	Dhaka	8,263	13,685	60.4	WHO DON598
Rajshahi	Rajshahi	3,747	5,832	64.2	WHO DON598
Chittagong	Chittagong	2,514	4,065	61.8	WHO DON598
Khulna	Khulna	1,568	2,337	67.1	WHO DON598
Jessore	Khulna	-	-	-	UNICEF (border

Nawabganj	Rajshahi	-	-	-	hotspot)
UNICEF (border hotspot)					

Data date: 14 April 2026 for WHO cases; 8 April 2026 for UNICEF hotspot designations.

District share = district cases / division total $\times$ 100.

Figure 5. District-Level Analysis Using WHO/UNICEF Data (April 2026)
WHO Disease Outbreak News DON598 + UNICEF Bangladesh Situation Report No.1

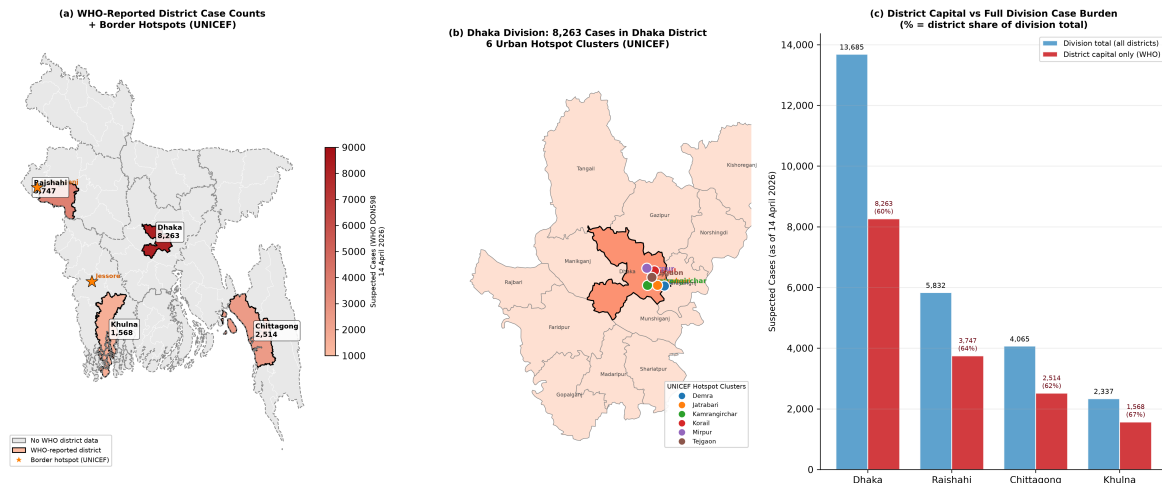


Figure 5. District-level analysis using WHO and UNICEF data (April 2026). Panel (a): national district map highlighting four WHO-reported districts (coloured by case count) and two UNICEF-designated border hotspots (stars); panel (b): Dhaka division zoom showing six UNICEF-identified urban transmission clusters within Dhaka district; panel (c): comparison of district capital case counts against full division totals, showing 60-67% concentration in district capitals.

4. Discussion

This study provides the first geospatial characterisation of the 2026 Bangladesh measles outbreak using formal geographic concentration metrics and temporal progression mapping. Our central finding is that the outbreak was highly spatially concentrated: Dhaka division alone accounted for nearly half of all suspected cases, and the Gini coefficient for confirmed cases (0.677) indicates a level of geographic inequality comparable to that observed in measles outbreaks in Ethiopia (Moran's $I = 0.154$) and Malaysia (significant spatial autocorrelation in Klang Valley) [9,10].

A key novel finding is the increase in HHI from 0.217 (mid-April) to 0.271 (early June), indicating that geographic concentration of the outbreak intensified over time. This pattern is inconsistent with a simple diffusion model where cases spread outward from an epicentre to surrounding areas. Instead, it suggests sustained focal transmission in Dhaka's high-density urban informal settlements - including Demra, Jatrabari, Kamrangirchar, and Mirpur - where population density, overcrowding, and accumulated immunity gaps create conditions for persistent measles transmission [5,6]. This pattern has been described as 'epidemiological connectivity' in high-density urban clusters, where spatial proximity amplifies transmission within clusters rather than between them [19].

The contrasting incidence profiles of Dhaka (108.0/100k, high density) and Barisal (93.9/100k, low

density but small population) highlight the importance of population-adjusted metrics alongside absolute case counts. Barisal's high incidence relative to its total case burden suggests a local outbreak with high attack rates, possibly linked to specific immunity gaps in peri-urban or rural communities in the division.

The null correlation between vaccination coverage and incidence ($r = 0.083$, $p = 0.845$) is consistent with our earlier report on this outbreak [18] and with findings from sub-Saharan Africa, where high national coverage coexists with spatially clustered populations of unvaccinated children [11]. The reactive MR campaign, launched on 5 April 2026, targeted 1.2 million children in 30 hotspot upazilas, yet the outbreak continued through June 2026, consistent with the limitations of reactive immunization once active transmission is established [2,8].

Several limitations merit acknowledgement. First, our analysis is at the division level ($n=8$), which precludes formal spatial autocorrelation statistics (Moran's I, LISA) that require larger observation sets [17]. Future analyses using district-level ($n=64$) or upazila-level ($n\sim 490$) data would enable more granular hotspot identification, as demonstrated for dengue in Bangladesh [13]. Second, surveillance data from DGHS press releases reflect reported cases and may underestimate true incidence in areas with limited healthcare access or surveillance capacity. Third, vaccination coverage data represent campaign targets and may not reflect actual immunity status among the at-risk population, particularly among zero-dose infants under nine months [6]. Fourth, no socioeconomic, demographic, or climate covariates were included in this ecological analysis; residual confounding cannot be excluded.

5. Conclusions

The 2026 Bangladesh measles outbreak was geographically highly concentrated, with Dhaka and Barisal divisions bearing disproportionate burden relative to their populations. The increasing Herfindahl-Hirschman Index over the outbreak's course indicates intensifying focal transmission rather than geographic diffusion, pointing to persistent transmission in high-density urban clusters. The null correlation between MR campaign coverage and incidence underscores that reactive immunization is insufficient to interrupt active transmission. These findings support targeted sub-divisional interventions - particularly in Dhaka's urban informal settlements - and call for strengthened routine EPI to reduce the accumulation of susceptible cohorts before future outbreaks occur. Geospatial surveillance at district and upazila levels should be integrated into Bangladesh's routine disease monitoring systems to enable earlier hotspot identification.

Declarations

Funding:

This research received no external funding.

Competing Interests:

The author declares no competing interests.

Data Availability:

All data and analysis scripts are publicly available at: <https://github.com/khalilurrahmanridoykhan/bangladesh-measles-2026>. Source data are publicly available from DGHS Bangladesh (<https://dghs.gov.bd>).

Ethics:

This study used only publicly available aggregated surveillance data. No individual patient data were accessed. Formal ethics review was not required.

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